

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460294.62713 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1237 +/- 0.0068
Transit depth (Rp/Rs)^2	1.53 +/- 0.17 [%]
Orbital Inclination (inc)	89.96 +/- 0.73
Airmass coefficient 1 (a1)	0.9319 +/- 0.0069
Airmass coefficient 2 (a2)	0.0485 +/- 0.0073
Scatter in the residuals of the lightcurve fit is	0.74 %
Best Comparison Star	#2 - [405, 338]
Optimal Aperture	5.54
Optimal Annulus	14.02
Transit Duration (day)	0.09027 +/- 0.00061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1237 ± 0.0068 vs 0.1301 (deviation 0.6σ)
Duration fit vs published	0.09027 d vs 0.0908 d (deviation 0.59σ)
Mid-time O–C	3.5 min
Depth SNR	11.6
Residual scatter	0.74 %

Light curve

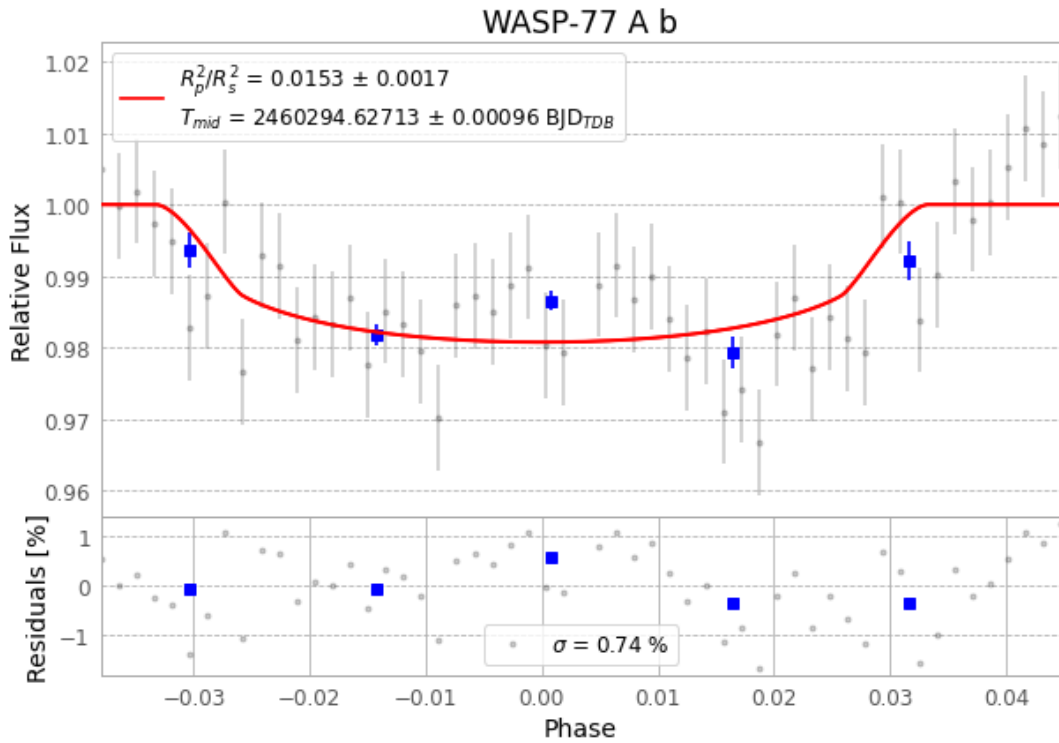


Figure 1 — FinalLightCurve_WASP-77 A b_2023-12-15.png (SHA-256 0fbf0ec0ab3df0c7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [512, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8e750002e6ddf5ae...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

55 FITS frames (36 MB), WASP-77231216014215.FITS → WASP-77231216042415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-231216.log (e7cd5fd2e83d...), FinalLightCurve_WASP-77 A b_2023-12-15.png (0fbf0ec0ab3d...), FinalLightCurve_WASP-77 A b_2023-12-15.csv (86611530f3e2...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 07:25Z.